CodeAlpha Project 2: Unemployment Analysis

Simple English • Beginner-friendly • Start-to-finish checklist

1. What is this project?

• Explore unemployment measurements over time, create charts, and describe trends.

• Important: a chart can show that a measure changed, but it does not by itself prove what caused the change.

2. What you need

• Use Google Colab. Upload `CodeAlpha_Unemployment_Analysis.ipynb` and run cells from top to bottom.

• The notebook tries to load a public CSV. If the link does not work, upload a suitable unemployment CSV and replace the URL with your uploaded filename.

3. Important words in simple English

• Trend = the general direction of values over time.

• Missing value = a blank or unavailable value.

• Data cleaning = making dates and numbers usable and removing unusable rows.

• Annual average = the average of available values in a year.

• Seasonality = a pattern that repeats regularly; enough regularly spaced data is needed to study it.

4. Start-to-finish steps

• Step 1: Open Colab and upload the notebook.

• Step 2: Load the CSV and print the column names. The names vary between datasets.

• Step 3: Check missing values and identify the date and unemployment-measure columns.

• Step 4: Convert dates to date format and measurements to numbers. Remove rows that cannot be used.

• Step 5: Create a line chart to see how the measure changes over time.

• Step 6: Print the average, minimum, maximum, and date range.

• Step 7: Group the data by year and plot annual averages.

• Step 8: Compare 2019 and 2020 only if both years are included and the data is comparable.

• Step 9: Write a conclusion using the chart and actual values printed by your run.

5. What to write in your conclusion

• State the country/region, units, and date range. Check these in the dataset source documentation.

• Describe whether the line mostly rises, falls, or fluctuates.

• Mention the years with the highest and lowest annual averages if appropriate.

• Discuss the 2019–2020 comparison cautiously. Do not claim COVID-19 caused a change based only on this chart.

• Mention limitations such as missing data, changes in definitions, and the dataset's geographic coverage.

6. New idea to improve it

• For monthly data covering several years, add a rolling 12-month average to smooth short-term fluctuations.

• If the dataset includes multiple regions, compare them only after checking that definitions and units match.

7. GitHub and submission

• Save the notebook as `Unemployment_Analysis.ipynb`.

• Create a README stating the data source, geography, units, date range, findings, and limitations.

• Use repository name `CodeAlpha_Unemployment_Analysis`.

• Run every cell and verify the charts before submitting.